



FAQs

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Frequently Asked Questions are a popular means of exchanging basic information, normally pointed to by RTFM or 'Read That Fine Manual' (amongst other more stressed meanings). Please send me any questions that you've been asked, or that you'd like an answer to anyway. If you have an answer already - please send both your Q and A!

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What is FLEX?

FLEX is the name of the Disk Operating System created by Technical Systems Consultants for the Motorola 6800 and 6809 microcomputer based computers. FLEX first arrived in 1977 with the 6800 version, later versions were for 8" and 5.25" disks and also for the 6809.

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What is the format of a FLEX disk? **NEW!**

(Contributed by Michael Evenson)

Track 0 does not have a sector 1. All of the other tracks don't have a sector 0. SWTPC Flex always had track 0 side 0 formatted as Single Density. Other manufacturers may have allowed track 0 side 0 to be Double Density. Gimix was one of these.

All of the sectors except the boot sector and sectors that belonged to random files were linked to the next logical sector by the first two bytes of each sector. This created a sector link chain. The chain was broken at the of the file by having the link set to 00 00. The start of the chain for each file is kept in the Directory entry for the file. Each linked sector also contains 2 bytes that signify its sequence number in the file. Flex uses this to determine the integrity of the file when it is read using the File Management System calls. This leaves 252 usable bytes in each sector to store the file.

This is the disk layout:

track 0	sector 0	boot sector
0	2	extended boot sector

0	3	System Information Record (SIR)
0	4	not used
0	5	start of the directory area.
...	...	the rest of track 0 is used for directory storage
1	1	start of the data and program storage area of diskette

On double sided diskettes, the sector numbers on the second side continue on from the numbers on the first side. That is, if track 1 sector 10 is the last sector on side 0, then track 1 side 1 will start with sector number 11 and end with sector number 20. Track 2 side 0 will start over with sector number 1.

This is the layout of the SIR:
The first 16 bytes are not used (all zeros)

offset(hex)	size(hex)	contents
-----	-----	-----
\$10	\$0B	Volume Label
\$1B	\$01	Volume Number High byte
\$1C	\$01	Volume Number Low byte
\$1D	\$01	First User Track
\$1E	\$01	First User Sector
\$1F	\$01	Last User Track
\$20	\$01	Last User Sector
\$21	\$01	Total Sectors High byte
\$22	\$01	Total Sectors Low byte
\$23	\$01	Creation Month
\$24	\$01	Creation Day
\$25	\$01	Creation Year
\$26	\$01	Max Track
\$27	\$01	Max Sector

The First User Track and Sector mark the beginning of the free chain. This is the chain of available unused sectors. The Last User Track and Sector mark the end of this chain.

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What is a random file?

Coming soon...

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What is the directory structure of a FLEX disk? **NEW!**

(Contributed by Michael Evenson)

Each Directory Entry is laid out as follows:
The first 16 bytes are not used (all zeros)

offset(hex)	size(hex)	contents
-----	-----	-----
\$10	\$08	File Name
\$18	\$03	File Extension
\$1B	\$02	Unused
\$1D	\$01	Start Track
\$1E	\$01	Start Sector
\$1F	\$01	End Track
\$20	\$01	End Sector
\$21	\$01	Total Sectors High byte
\$22	\$01	Total Sectors Low byte
\$23	\$01	Random File Indicator

\$24	\$01	Unused
\$25	\$01	Creation Month
\$26	\$01	Creation Day
\$27	\$01	Creation Year

The Start Track and Sector mark the beginning of the linked list of sectors that make up the file. The directory sectors are linked, but do not contain a sequence number like other linked sectors do.

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What are DSK files?

DSK files, or files with the file name extension DSK, are file representations of a FLEX disk. The complete disk structure is copied, the only difference being that the inter-sector and inter-track bit strings used by the floppy disk controllers are not present.

FLEX DSK files are a very useful way of saving the contents of aging floppy disks. Also, and fortunately, all the FLEX software emulators use the DSK format as their basic FLEX disk format. Using DSK files then becomes a true representation of the floppy-based age for the FLEX Disk Operating System.

Software on this website and CD is stored in the DSK format. Tools are available to extract and insert FLEX files into the FLEX DSK images.

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What is the difference between 6800 and 6809 FLEX?

Coming soon...

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What is UniFlex?

UniFLEX was the last generation of the TSC FLEX Operating Systems. UniFLEX was a true multi-user unix-like operating system running on the 8/16 bit 6809 microprocessor. Unfortunately for TSC the development of UniFLEX coincided with the introduction of the IBM PC Jr and the 8-bit 8088 and then the 16-bit 8086, so as the market fell for 8-bits bad, 16-bits better, UniFLEX saw its market share disappear and UniFLEX was sold to an investor and then faded from view.

Fortunately Randy Lewis found one of the master tapes for UniFLEX and has put all the UniFLEX source files onto his website (see Links). Download these files to see just what an accomplishment was made in putting unix onto an 8-bit machine.

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